

CCY2001 – Introduction to Cybersecurity

Lecture 4 – Securing Your Computer

Dr. Hany Elshall

CCIT

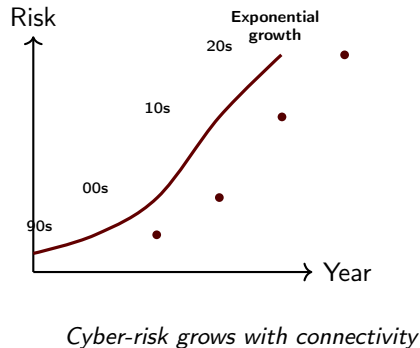
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Lecture Roadmap

- 1 Introduction
- 2 Firewalls
- 3 Antivirus Software
- 4 Anti-Spyware & Anti-Spam
- 5 Security Updates & Secure Browsing
- 6 Device Scanning & Social Engineering
- 7 Miscellaneous Security Tips
- 8 Summary & Takeaways

Why Does Computer Security Matter?

- Computers are **indispensable** in every aspect of modern life
- The Internet increased global dependency across *all* sectors
- Greater dependency $\Rightarrow$ greater **security risks**
- Early malicious programs caused **major financial losses**, revealing the urgent need for security



Types of Computer Threats

PHYS Physical Threats

Overt – visible, direct damage
(theft, fire, vandalism)

Covert – hidden, indirect damage
(hardware tampering, surveillance devices)

CYBER Cybersecurity Threats

Computer Security

Protecting individual systems from unauthorised access

Network Security

Safeguarding communication networks

Data Security

Protecting digital information during storage & transmission

Key Insight: Modern threats are often *blended* – attackers combine physical and cyber methods to bypass layered defences.

Escalation of Cybercrime

- Cybercrime has grown in **scale**, **sophistication**, and **financial impact**
- **Cyber terrorism** targets critical infrastructure, government systems, and public services
- Attackers exploit every new technology advancement
- Comprehensive cybersecurity requires:
 - Physical & electronic protection
 - Continuous monitoring
 - Early detection
 - Proactive defence strategies

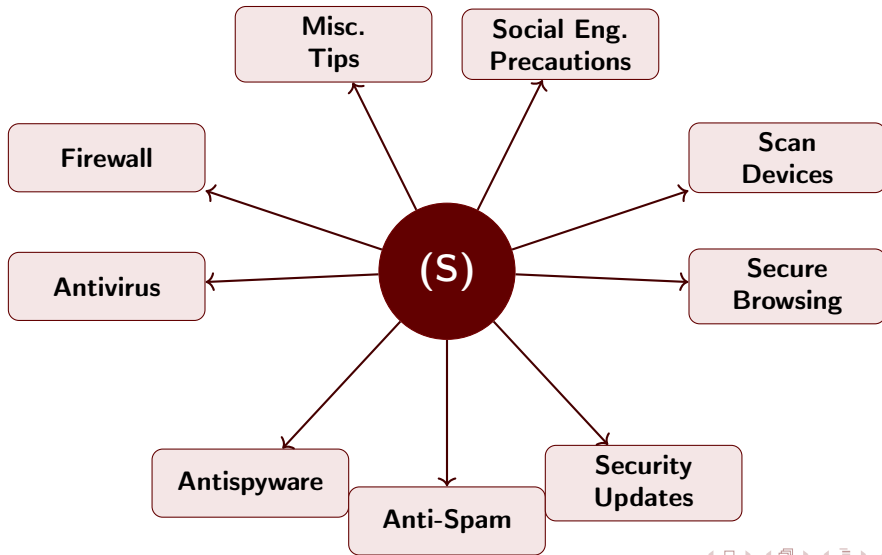
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Global cybercrime cost in 2023

2,200+

Cyberattacks per day worldwide

Major Components for Securing a Computer



What is a Firewall?

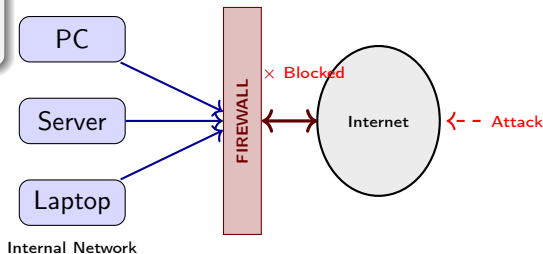
Definition

A **firewall** is a security system that **monitors and controls** incoming and outgoing network traffic based on *predetermined security rules*.

It establishes a barrier between:

- 🔒 A **trusted internal network** (e.g. corporate LAN)
- ! An **untrusted external network** (e.g. the Internet)

Think of it as a **security guard at a building entrance** – checking IDs before anyone gets in or out.



How a Firewall Works

- 1 **Monitor traffic** – inspects every data packet entering/leaving the network
- 2 **Apply rules** – checks packet against rules (IP address, port, protocol)
- 3 **Block or Allow** – passes safe traffic; drops/rejects threats
- 4 **Filter both ways** – protects inbound *and* outbound traffic

Default Action – Important!

A firewall needs a **default policy** for traffic not covered by any rule.

Best practice: Set default to DROP or REJECT

This means: “if not explicitly permitted, it is denied.”

Example: SSH Rule

ALLOW SSH from 10.0.0.5
DROP SSH from any other

Importance of Firewalls

- ✓ **Prevent Unauthorised Access**
Like a locked door with a guard
- ✓ **Block Malicious Traffic**
Viruses, phishing, DoS attacks stopped before entry
- ✓ **Protect Sensitive Information**
Prevents data theft or accidental leaks

- ✓ **Control Network Usage**
Parental controls, workplace policies, filtering
- ✓ **Mitigate Insider Risks**
Detects suspicious apps or data exfiltration from within
- ✓ **Data Loss Prevention (DLP)**
Integrates with tools that detect sensitive content in traffic

Firewall Types – Across Network Layers

Type	Description
Network Layer	Filters by source/destination IP, protocol, routing info
Transport Layer	Filters by TCP/UDP port numbers and connection states
Application Layer	Deep inspection of application-level protocols (HTTP, FTP)
Host-Based	Installed on individual devices – protects that device only
NAT Firewall	Masks internal IPs; blocks all unsolicited inbound traffic

Remember: Defence-in-depth uses *multiple* firewall types together rather than relying on one.

Packet-Filtering

Examines individual packets – IP, port, protocol.
Fast but stateless; can be tricked.

Stateful Inspection

Tracks *connection states* – knows if a packet belongs to an established session. Much smarter than simple packet filtering.

Proxy / App-Level Gateway

Acts as *intermediary* – receives requests, inspects them, then forwards. Provides application awareness.

Next-Generation Firewall (NGFW)

Best of all worlds:

- ✓ Intrusion Prevention (IPS)
- ✓ Deep Packet Inspection (DPI)
- ✓ Application awareness
- ✓ Threat intelligence feeds

Firewall in Action – Worked Example

Scenario: A company web server is publicly reachable.

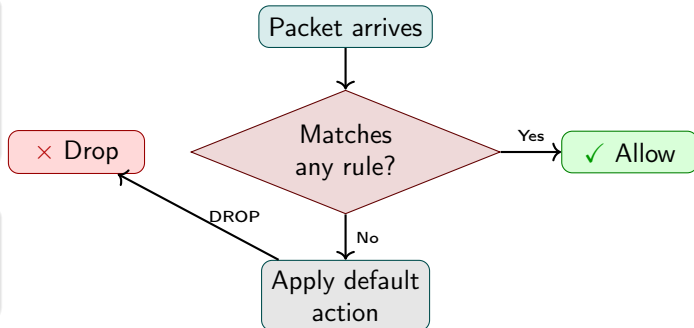
Firewall Rules Configured

```
ALLOW HTTP (port 80) from any
ALLOW HTTPS (port 443) from any
ALLOW SSH (port 22) from 10.0.0.5
DROP all other traffic
```

Attack Attempt

Attacker IP: 203.0.113.77
Target: 192.168.1.10:22 (SSH)

Result: SSH from 203.0.113.77 is **DROPPED** – not in the allow-list.



Common Firewall Bypass Techniques

- ! **Malware / Payload Delivery**

Exploiting unpatched firewall vulnerabilities

- ! **Phishing Attacks**

Stealing admin credentials to modify rules

- ! **Social Engineering**

Manipulating staff to reveal access credentials

- ! **Deep-Packet Inspection Evasion**

Hiding payloads inside encrypted traffic

- ! **VPN / Proxy Abuse**

Tunnelling malicious traffic around firewall rules

- ! **IPS Evasion**

Crafting traffic so that IPS signatures do not trigger

A firewall is one layer in defence-in-depth – it is not a silver bullet!

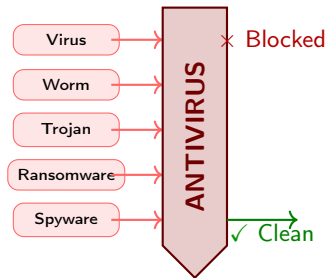
What is Antivirus Software?

Definition

Software designed to **prevent, detect, and remove malware** from a computer or device.

Core capabilities:

- Scans files against a **virus signature database**
- Monitors system behaviour for **suspicious activity**
- Provides **real-time protection** against viruses, worms, Trojans, ransomware
- Can **remove, delete, or quarantine** infected files
- Modern solutions add **cloud intelligence & behaviour analysis**



How Antivirus Works

Scan Scanning

Checks files against a database of known virus **signatures**

AI Behavioural

Identifies **suspicious patterns** even for unknown malware

Alert Real-Time

Runs in background continuously, catches threats **as they happen**

Cloud Cloud Intel

Leverages global threat databases to neutralise **emerging** threats

Types of Antivirus Software

Type	How it works	Weakness
Signature-Based	Matches files to known virus signatures	New/modified malware
Heuristic-Based	Looks for patterns resembling known malware	False positives
Behaviour-Based	Watches <i>how</i> programs act at runtime	May react slowly
Cloud-Based	Offloads analysis to cloud servers	Needs internet
AI/ML-Based	Predicts threats from learned patterns	Large data needed
Network/Gateway	Scans traffic at the network boundary	Misses encrypted flows

Best Practices for Antivirus

- ✓ **Enable at startup** – ensure AV starts automatically
- ✓ **Keep it running** – never disable real-time protection
- ✓ **Update regularly** – virus definitions must stay current
- ✓ **Run quick scans often** – detect immediate threats
- ✓ **Run full scans** – every few weeks for deep protection
- ✓ **Scan external media** – always scan USB/CD before use
- ! **Never ignore warnings** – every alert is worth investigating

Antivirus vs. Firewall – Quick Comparison

Firewall controls *what traffic can enter/leave* your network.

Antivirus controls *what files/programs run* on your device.

Both are needed – they complement each other.

What is Spyware?

Malicious software that **secretly collects information** from a device – keystrokes, browsing habits, passwords, financial data – without the user's knowledge.

Anti-Spyware detects, prevents, and removes spyware using:

- **Signature-based detection** – matching known spyware patterns
- **Rules-based detection** – monitoring for suspicious behaviour

Examples: SpyHunter, Spybot, Malwarebytes

Best Practices

- Use **dedicated** anti-spyware tools
- Keep software **updated**
- **Scan regularly**
- Avoid **unfamiliar websites**
- Close pop-ups via the **X button** only
- **Block pop-ups** in browser settings
- Never open **unknown email attachments**

Anti-Spam Software

Spam is one of the primary vectors for spreading malware.

Brief History

- First spam email $\approx$ **1978**
- By mid-1990s, spam reduced company productivity significantly
- Led to the creation of **MAPS** (Mail Abuse Prevention System)
- Today: Gmail, Yahoo, Outlook filter **99%+** automatically

ISP-level filtering alone is not sufficient – add **personal anti-spam** tools within your email application.

How Anti-Spam Works

- ➊ **Content filtering** – scans for spam keywords (“Free”, “Millionaire”)
- ➋ **Header analysis** – checks sender IP & routing metadata
- ➌ **Blacklist filtering** – known spammer IPs are blocked
- ➍ **Rule-based filtering** – custom rules per sender / attachment
- ➎ **Machine learning** – learns from user feedback patterns

Security Updates – Why They Matter

What are Security Updates?

Patches released by developers to **fix known vulnerabilities** before attackers can exploit them.

Security updates:

- ✓ **Patch vulnerabilities** – close doors before hackers can enter
- ✓ **Protect against new malware** – add defences against latest threats
- ✓ **Prevent data breaches** – reduce risk of sensitive data theft
- ✓ **Ensure stability** – fix bugs that could cause crashes
- ✓ **Maintain compliance** – meet regulatory requirements

Best Practices

- **Install promptly** as updates are released
- **Enable automatic updates** – especially for mobile
- **Restart when prompted** – some updates finalise only after reboot
- **! Beware fake updates!**
Never install from pop-ups or emails – only from *official sources*

Secure Browser Settings

Why Browser Security Matters

- Browsers handle **passwords, credit cards, personal history**
- Hackers exploit browser vulnerabilities to inject malware
- Phishing sites can trick users into giving away credentials

Browser Security Features

- ✓ Pop-up blockers
- ✓ Anti-malware filters
- ✓ Safe browsing warnings
- ✓ HTTPS enforcement
- ✓ Sandboxed tabs

Browser Best Practices

- **Update browser** immediately when prompted
- **Strong, unique passwords** + password manager
- **Disable autofill** for passwords and payment info
- **Clear cache & cookies** regularly
- Only install **trusted extensions**; remove unused ones
- Use **private/incognito** mode when on shared devices
- Browse with a **standard user account** (not admin)

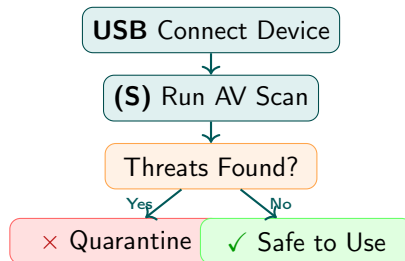
Scanning Data Transfer Devices

Risk: External devices are a major carrier of malware.

- Flash drives, CDs, DVDs, external hard drives can carry viruses, worms, Trojans
- **Even brand-new devices** can be pre-infected (supply chain attacks)

Correct Procedure

- 1 **Plug in** the external device
- 2 **Scan immediately** with updated AV software
- 3 Wait for the **all-clear** result
- 4 **Only then** copy or open files



Social Engineering Attacks

Definition

Social engineering manipulates *people* rather than technology to gain unauthorised access, using psychology and deception.

How it works:

- ➊ **Research** – gather info on the target to build credibility
- ➋ **Establish trust** – impersonate IT staff, bank, or official
- ➌ **Manipulate** – trick victim into clicking links, revealing passwords, or transferring funds

Attack channels: email, SMS, phone calls, social media, chat platforms

Common Attacks

Phishing – fake emails mimicking trusted brands

Vishing – phone calls from “tech support”

Smishing – SMS messages with malicious links

Pretexting – fabricated scenario to extract info

Baiting – leaving infected USB drives in public

The “Duck Test” – Trust Your Instincts

“If it looks like a duck, swims like a duck, and quacks like a duck – then it probably is a duck.”

Applied to cybersecurity:

- If something **feels like a scam** – it probably is
- Do **not** provide confidential information, regardless of:
 - How urgent the request seems
 - How great the promised reward is
 - How harmless it may appear
- Social engineering works by targeting **people who follow rules**

Quick Checklist

- ✓ Did you expect this email/call?
- ✓ Does the sender address look correct?
- ✓ Are you being pressured to act *immediately*?
- ✓ Does the request bypass normal procedures?
- ✓ Is there an unusual reward or threat?
- ! If **any** answer is suspicious – **stop and verify** through a trusted channel.

Miscellaneous Security Tips (1/2)

Browsing & Software

- ✓ Avoid untrusted or faulty websites
- ✓ Set browser to high security and privacy
- ✓ Uninstall all unused applications
- ✓ Never install unknown applications
- ✓ Only install necessary plugins
- ✓ Prefer paid/reputable apps over free unknown ones
- ✓ Keep all apps and OS updated to latest versions

User Accounts

- ✓ Always create **separate user accounts** per person
- ✓ Use a **guest account** with minimum privileges
- ✓ Use a **standard** account for daily tasks (not admin)
- ✓ Use **strong, unique passwords** for every account
- ! **Never reuse** passwords across different services

Miscellaneous Security Tips (2/2)

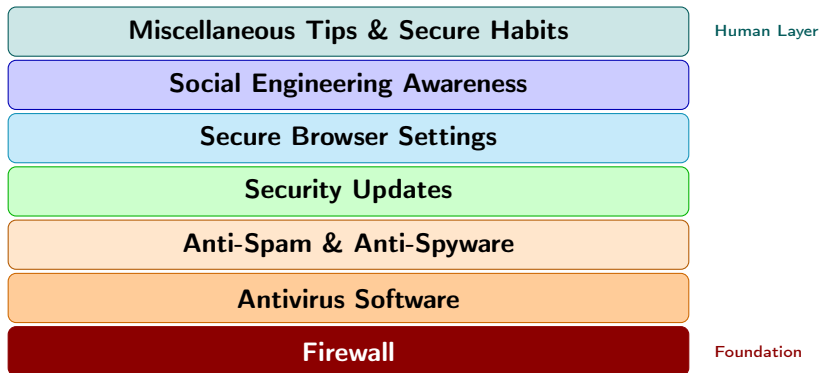
System & Data

- ✓ Never skip security patches or updates
- ✓ Check for security updates **daily**
- ✓ Turn off your computer when not in use
- ✓ Run scheduled virus scans **regularly**
- ✓ Create system **restore points / checkpoints**
- ✓ Always **back up** valuable data

Downloads & Data Sharing

- ✓ Only download from **reliable, official** websites
- ✗ Never use **public data-sharing** sites for sensitive data
- ✓ Use **secure, reputable platforms** for file transfers
- ✓ **Encrypt** important files (built-in OS tools or third-party)
- ✓ Take security as a **serious, ongoing** responsibility

Putting It All Together – The Security Stack



Security is **layered**: each component addresses different threats. All layers work together – removing one weakens the whole stack.

Key Takeaways

- ➊ **Threats are diverse** – physical, network, data, and human-based
- ➋ **Firewalls** are the first line of defence – control what enters/leaves
- ➌ **Antivirus** detects and removes malware from your device
- ➍ **Anti-Spyware/Spam** close gaps that AV may miss
- ➎ **Updates** are your fastest free security upgrade – do not delay
- ➏ **Secure your browser** – it handles your most sensitive data
- ➐ **Scan all external media** before use
- ➑ **Social engineering** targets people, not computers – stay alert and *trust your instincts*

Thank You!

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Questions? & Discussion

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